Analyzing Representation Dynamics in a Masked Joint-Embedding Predictive Architecture for Audio

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Abstract

We study the representation dynamics of a masked joint-embedding predictive architecture (JEPA) trained on piano audio. Rather than optimizing for downstream discrimination, the model is trained using a predictive self-supervised objective and evaluated through representation diagnostics, including temporal consistency, isotropy, and piece-level geometry. Representations are analyzed across multiple training checkpoints, with downstream classification used only as an auxiliary probe. Our results provide an empirical characterization of temporal structure and embedding geometry under predictive training with variance regularization, within computational constraints.

1 Introduction

Self-supervised representation learning has become a central approach for modeling high-dimensional data without explicit labels. In audio and vision, recent methods increasingly favor predictive objectives over contrastive discrimination, aiming to learn representations by modeling structure and continuity rather than class separation. Joint-embedding predictive architectures (JEPAs) exemplify this direction by training models to predict latent representations of masked or future content instead of reconstructing inputs or contrasting positive and negative pairs.

Despite strong downstream performance reported for JEPAs, comparatively little attention has been paid to how their internal representations evolve over training. In particular, it remains unclear how predictive learning and variance regularization jointly shape temporal consistency, effective dimensionality, and inter-sample similarity in the learned embedding space. This question is especially important in audio, where temporal continuity is a dominant organizing principle of the data.

In this work, we present an empirical study of a masked joint-embedding predictive architecture trained on piano audio. Our goal is not to introduce a new method, but to examine how predictive self-supervised training shapes representation structure over the course of training. We analyze representations extracted at multiple checkpoints using simple diagnostics that probe temporal consistency and global geometry, while using downstream classification only as an auxiliary evaluation.

We observe several consistent patterns across training. Representations exhibit strong temporal smoothness early on, which becomes increasingly localized as training progresses. At the same time, the embedding space becomes more isotropic and expands in effective dimensionality, while separation between musical pieces decreases. Downstream classification performance peaks at intermediate stages and does not improve monotonically with training.

Together, these findings suggest that, in our setting, predictive JEPA-style training favors temporal coherence and predictability relative to discriminative separation.

2 Related Work

Predictive self-supervised learning and JEPA. Joint-Embedding Predictive Architectures (JEPAs) frame self-supervised learning as *feature prediction* in representation space rather than reconstruction or contrastive discrimination. I-JEPA introduced this paradigm for images by predicting representations of masked target blocks from a context region using a student encoder, an EMA teacher, and a lightweight predictor, avoiding heavy handcrafted augmentations [1]. For video, V-JEPA revisited feature prediction as a standalone objective and showed that predicting video features yields representations transferable across motion and appearance tasks using frozen backbones [4]. V-JEPA 2 further scales this direction with internet-scale video data and investigates extensions toward prediction/planning and action-conditioned

modeling [2].

JEPA for audio. Several works adapt JEPA-style objectives to audio by operating on time–frequency patches of spectrograms. A-JEPA presents a JEPA formulation for audio spectrograms and emphasizes masking strategies that respect time–frequency structure [6]. More recently, WavJEPA explores JEPA-style pretraining directly on raw waveforms, targeting robust general-purpose audio representations without relying on spectrogram tokenization [8]. These efforts motivate studying not only downstream performance but also how predictive training shapes *temporal behavior* and representation geometry in audio domains.

Regularization and representation geometry. A recurring challenge in non-contrastive and predictive SSL is preventing collapse and encouraging well-conditioned embeddings. LeJEPA provides a theoretical account arguing that an isotropic Gaussian embedding distribution is risk-optimal for downstream prediction, and introduces *Sketched Isotropic Gaussian Regularization (SIGReg)* to push learned embeddings toward that distribution [3]. In a related spirit for video JEPA, VJ-VCR adds variance–covariance regularization to stabilize training and avoid degenerate representations [5]. Our work is aligned with these perspectives: we train a masked audio JEPA with SIGReg-inspired regularization and focus on *diagnostics* (temporal consistency, isotropy proxies, effective dimensionality, and intra-/inter-piece geometry) across checkpoints, rather than treating downstream classification as the sole objective.

Dataset. We train and evaluate on MAESTRO, a large-scale dataset of aligned piano audio and MIDI performances [7].

3 Method

3.1 Dataset and Preprocessing

We train and evaluate our model on a subset of the MAESTRO v3.0.0 dataset, consisting of solo piano performances. Audio recordings are converted to log-mel spectrograms and segmented into fixed-length windows of W frames. All windows are extracted using identical preprocessing parameters during training and evaluation to ensure representational consistency.

Each spectrogram window is treated as a two-dimensional time-frequency array and partitioned into a fixed grid of non-overlapping patches. These patches form the units processed by the encoder. Windows shorter than W frames are zero-padded. No global normalization across windows or pieces is applied beyond standard log-mel computation.

During training, masking is applied at the patch level. A random subset of patches is masked, yielding a partially observed input for the context encoder. The corresponding unmasked window is provided to the target encoder. Masking patterns are sampled dynamically and shared across the batch for stability.

All downstream evaluation and representation diagnostics reuse the same windowing and patchification scheme, without masking.

3.2 MaJEPA Architecture

Our model follows a masked joint-embedding predictive architecture (JEPA) adapted to time–frequency audio representations, with additional variance regularization inspired by LeJEPA.

The input to the model is a log-mel spectrogram window, partitioned into a fixed grid of non-overlapping time–frequency patches. Each patch is linearly projected into a D-dimensional embedding, yielding a sequence of patch tokens arranged on a two-dimensional grid. No special classification token is used; all representations are derived from patch tokens alone.

The architecture consists of two encoders with identical Vision Transformer (ViT) structure: a *context encoder* (student) and a *target encoder* (teacher). The context encoder processes a partially masked input, while the target encoder processes the corresponding unmasked input. The target encoder parameters are updated as an exponential moving average (EMA) of the context encoder parameters and are not directly optimized by gradient descent.

Both encoders produce grid-structured latent patch representations. For the context encoder, only representations at unmasked patch locations are retained. These context representations are augmented with positional encodings and passed to a lightweight predictor network, which outputs predicted embeddings aligned with the target encoder's representations at masked locations.

In addition to the predictive objective, the model applies a distributional regularization term to the latent representations produced by the target encoder. This regularization biases the aggregated latent distribution toward an isotropic Gaussian and is implemented using a characteristic-function-based statistic, as described in Section 3.3.

3.3 Training Objective

Training optimizes a combination of a masked joint-embedding predictive loss and a variance regularization term applied in representation space.

Let x denote a log-mel spectrogram window partitioned into a grid of patches. A random subset of patch indices $\mathcal{M}$ is selected for masking, yielding a partially observed input $\tilde{x}$. The context encoder f_θ processes $\tilde{x}$, while the target encoder $f_{\bar{\theta}}$, parameterized as an exponential moving average of θ, processes the unmasked input x. Gradients are not propagated through the target encoder.

For each masked patch index $i \in \mathcal{M}$, a lightweight predictor network produces a predicted embedding $\hat{z}_i$, which is trained to match the corresponding target encoder embedding z_i. The predictive loss is defined as:

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\hat{z}_i - z_i\|^2.$$

To regularize the geometry of the latent space, we additionally apply a SIGReg loss to the target encoder representations. Following LeJEPA, SIGReg measures the deviation of the empirical characteristic function of randomly projected latent embeddings from that of a standard normal distribution. The loss is computed on a subsampled set of latent vectors, after centering and per-dimension standardization, and is averaged over multiple random projections.

The final training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \, \mathcal{L}_{\text{SIGReg}},$$

where λ controls the strength of the regularization. SIGReg is applied throughout training and uses step-dependent random projections seeded by the global training step.

3.4 Representation Extraction

All representations used for evaluation are derived from the trained MaJEPA (Maestro JEPA) model described in Sections 3.2 and 3.3. In all cases, embeddings are extracted from the EMA (target) encoder, which is used as a fixed feature extractor. During extraction, no masking is applied and the model operates in evaluation mode.

For a given input window, the target encoder produces a grid of patch-level latent representations corresponding to the input time–frequency patches. These patch representations constitute the fundamental output of the model. To obtain a fixed-dimensional embedding per window, we apply mean pooling over all patch tokens. No special classification token or learned pooling mechanism is used.

Because the model is trained on windowed audio, representation extraction additionally requires defining how window-level embeddings are sampled and aggregated across each musical piece. We use three related extraction methods, each corresponding to a different evaluation objective.

3.4.1 Piece-Level Embeddings

For downstream evaluation tasks, including linear probing and k-nearest neighbors classification, we construct a single embedding per piece.

For each musical piece, we sample 20 windows using random temporal offsets. Each window is independently encoded by the target encoder and pooled to produce a window-level embedding. The final piece representation is defined as the mean of these window embeddings:

$$z_p = \frac{1}{20} \sum_{i=1}^{20} z_{p,i}.$$

This aggregation enforces invariance to window position by construction and substantially reduces computational and storage requirements. Only these piece-level embeddings are used for downstream classification experiments.

3.4.2 Window-Level Geometry Embeddings

To analyze the geometry of the learned representation space independently of piece-level aggregation, we extract window-level embeddings without averaging across windows.

For each piece, we extract 50 windows by repeatedly sampling a single random window per pass over the dataset, using different random seeds. Each window is encoded independently, yielding a set of pooled window embeddings. Metadata identifying the source piece and window start location is retained. These embeddings are used exclusively for representation diagnostics, including isotropy, effective dimensionality, and intra- versus inter-piece similarity, and are never used for classification.

This procedure approximates broad temporal coverage while avoiding the cost of exhaustively extracting all possible windows.

3.4.3 Temporal Embeddings

To study temporal consistency, we extract embeddings at controlled temporal offsets within each piece.

For each piece, we randomly select three center windows. Around each center, we extract embeddings at temporal lags $\ell \in \{-10, \ldots, 10\}$ with a fixed stride

of 10 frames. Window start positions are deterministically forced to ensure temporal alignment. Each window is encoded independently and pooled to a fixed-dimensional vector.

This yields temporally ordered embedding sequences suitable for measuring how representation similarity varies as a function of temporal separation. The same extraction protocol is applied at multiple training checkpoints (10k, 25k, 50k, 75k, and 100k steps) using identical parameters and random seeds to ensure comparability across training time.

Compute considerations. All extraction protocols and aggregation choices are guided by practical computational constraints. The final model was trained for approximately nine hours to reach 100k steps on a single NVIDIA T4 GPU, and exhaustive window- or token-level extraction would have been prohibitively expensive in both time and storage. We therefore rely on sparse but structured sampling strategies that provide broad temporal and piece-level coverage while remaining tractable. More exhaustive alternatives—such as denser temporal trajectories, token-level analyses, or learned pooling mechanisms—may yield additional insight, but are left to future work.

4 Results

4.1 Temporal Consistency

Figure 1 shows mean cosine similarity between window-level embeddings as a function of temporal lag across training checkpoints. At all checkpoints, similarity is highest at zero lag and decreases monotonically with increasing temporal offset.

Early in training, similarity decays gradually, indicating broad temporal invariance. With continued training, the decay becomes sharper, yielding stronger sensitivity to small temporal offsets while preserving local continuity. This pattern is consistent with increasingly localized temporal sensitivity over training, rather than uniform invariance across time.

4.2 Downstream Classification Performance

We evaluate linear probing and k-nearest neighbors (kNN) classification on piece-level embeddings as auxiliary diagnostics. As shown in Figure 2, linear probe accuracy improves rapidly early in training, peaks around 75k steps, and slightly decreases thereafter. kNN accuracy follows a similar trend and stabilizes for moderate values of k.

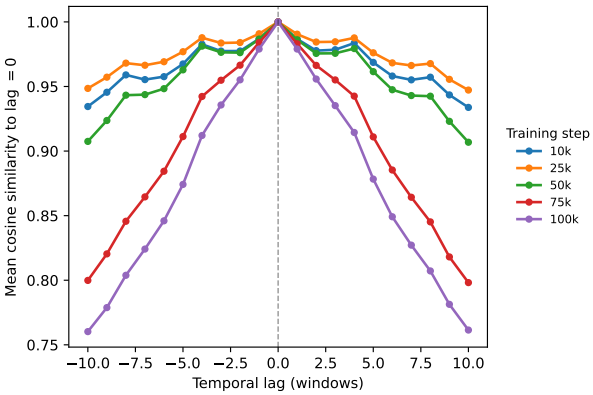


Figure 1: Mean cosine similarity between window-level embeddings as a function of temporal lag. Curves correspond to different training checkpoints, illustrating progressively sharper temporal localization with training.

Notably, downstream accuracy does not increase monotonically with training and does not track later improvements in isotropy or effective dimensionality, reinforcing that discriminative performance is not directly optimized by the training objective.

4.3 Embedding Norm Dynamics

Figure 3 illustrates the distribution of window-level embedding norms across training checkpoints. Early representations exhibit large norms with broad distributions. As training progresses, the distributions contract and concentrate around smaller values.

This contraction is consistent with the effect of variance regularization, which discourages anisotropic scaling and stabilizes representation magnitude. Importantly, this normalization occurs alongside increased isotropy and effective dimensionality rather than representational collapse.

4.4 Isotropy and Effective Dimensionality

We quantify isotropy using the mean cosine similarity between randomly sampled embedding pairs and estimate effective dimensionality via the participation ratio of the covariance spectrum (Figure 4). Over training, random-pair cosine similarity decreases toward zero, indicating increasingly isotropic representations.

Concurrently, the participation ratio increases steadily, suggesting that representations expand to occupy a higher-dimensional subspace. Together, these

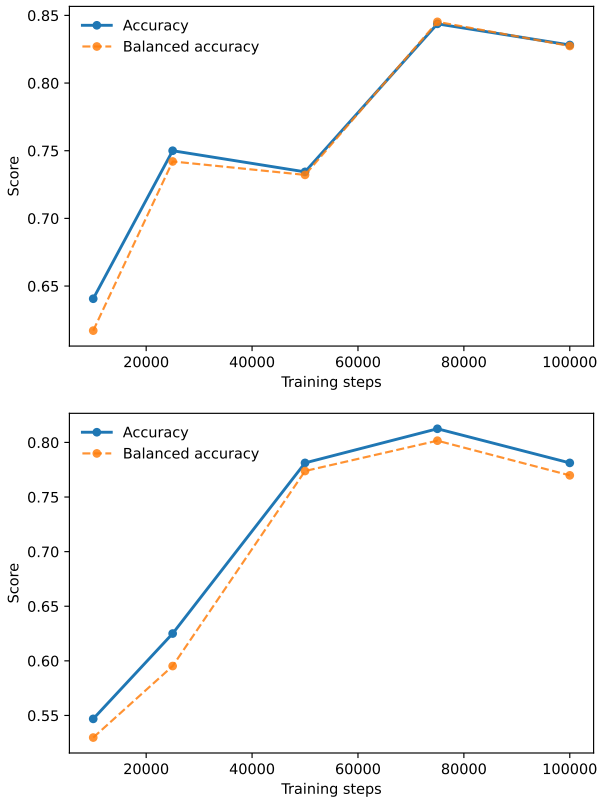


Figure 2: Auxiliary downstream diagnostics on piece-level embeddings. *Top*: linear probe accuracy and balanced accuracy versus training step. *Bottom*: k-nearest neighbors accuracy at fixed $k = 10$. Performance peaks at intermediate training stages.

trends indicate that, in our experiments, variance regularization is associated with increased uniformity and effective dimensionality.

4.5 Intra- vs Inter-Piece Geometry

Figure 5 compares cosine distance distributions for window pairs drawn from the same piece (intra-piece) versus different pieces (inter-piece). Early checkpoints exhibit partial separation between these distributions, indicating some piece-level clustering.

As training progresses, the distributions increasingly overlap. In this setting, representations become less separated by piece identity as training progresses, despite improvements in isotropy and temporal consistency.

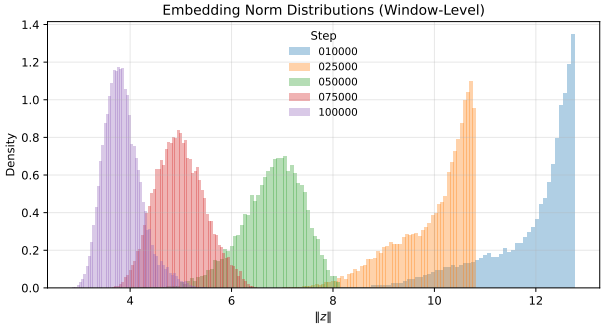


Figure 3: Distribution of window-level embedding norms across training checkpoints. Norms contract and stabilize over training, consistent with variance regularization.

4.6 Geometry–Isotropy Relationship

Figures 6 relate piece-level separation to training progress and isotropy. As representations become more isotropic and higher-dimensional, the difference between mean inter- and intra-piece cosine distances decreases monotonically.

We observe that as representations become more isotropic, piece-level separation decreases.

5 Discussion

This work presents an empirical analysis of representation dynamics in a masked joint-embedding predictive architecture trained on piano audio. By evaluating temporal consistency, representation geometry, isotropy, and separability across training checkpoints, we provide a detailed view of how predictive objectives and variance regularization are reflected in the learned embedding space over training. Rather than focusing on peak downstream performance, our analysis highlights the structural properties induced by JEPA-style training.

A central observation is the progressive sharpening of temporal representations. Early in training, embeddings exhibit broad temporal invariance, with high similarity across wide temporal offsets. As training proceeds, this invariance becomes increasingly localized, yielding representations that preserve short-range continuity while becoming more sensitive to fine-grained temporal differences. This behavior is consistent with the predictive nature of the objective: the model learns to encode information that is useful for predicting masked content, which naturally emphasizes local temporal structure without enforcing global invariance.

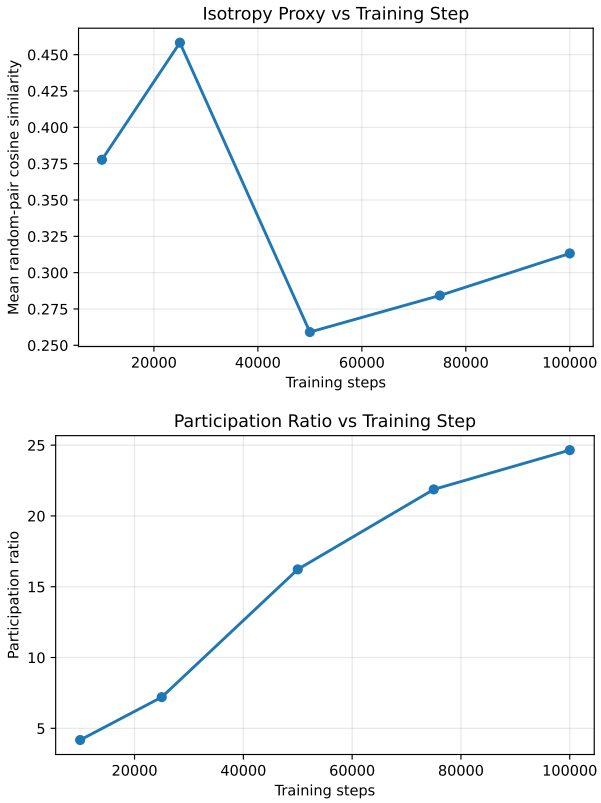


Figure 4: Global representation geometry across training. *Top*: isotropy proxy measured by mean cosine similarity of random embedding pairs. *Bottom*: participation ratio of the covariance spectrum, indicating increasing effective dimensionality.

We also observe a consistent increase in isotropy and effective dimensionality over training, accompanied by a contraction of embedding norms. These trends are consistent with the role of SIGReg regularization, which encourages embeddings to approximate an isotropic Gaussian distribution. Importantly, this regularization does not lead to representational collapse; instead, embeddings expand to occupy higher-dimensional subspaces while remaining well-conditioned. This suggests that variance regularization acts as a geometric constraint that stabilizes representation space rather than restricting expressivity.

At the same time, increased isotropy is accompanied by a reduction in piece-level separability. Both the overlap of intra- and inter-piece distance distributions and the monotonic decrease in a separation proxy indicate that predictive training does not prioritize clustering by semantic identity. Downstream classification performance peaks at intermediate training stages and does not track later improvements in isotropy or dimensionality. Together, these results illustrate an empirical tension between global uniformity, temporal coherence, and discriminative separation in this setting.

Several limitations of this study should be emphasized. First, all experiments are conducted on a restricted subset of the MAESTRO dataset, focusing on piano performances by only two composers. While this choice enables controlled analysis and tractable computation, it limits the generality of our conclusions. Representation dynamics may differ substantially for other instruments, musical styles, or broader composer distributions.

Second, our analysis is based on a single architecture and training configuration, evaluated under a fixed masking strategy and variance regularization strength. Although we observe consistent trends across training checkpoints, we do not compare against alternative self-supervised objectives, contrastive baselines, or ablations of the regularization scheme. As a result, our findings should be interpreted as descriptive rather than comparative.

Third, representation extraction relies on sparse but structured window sampling due to computational constraints. The final model was trained for approximately nine hours to reach 100k steps on a single NVIDIA T4 GPU, and more exhaustive window- or token-level analyses were not feasible. While our sampling strategy captures broad temporal and piece-level behavior, finer-grained analyses may reveal additional structure not captured here.

Finally, downstream classification is used only as an auxiliary probe and should not be interpreted as a comprehensive measure of representation quality. Our results highlight correlations between geometric properties and downstream performance, but do not establish causal relationships.

We emphasize that our findings are descriptive and specific to this setting, and are not intended as universal claims about JEPA training.

6 Conclusion

We presented an empirical study of representation dynamics in a masked joint-embedding predictive architecture trained on piano audio. By analyzing temporal consistency, representation geometry, isotropy, effective dimensionality, and piece-level separation across training checkpoints, we characterized how predictive objectives and variance regularization jointly shape the learned embedding space.

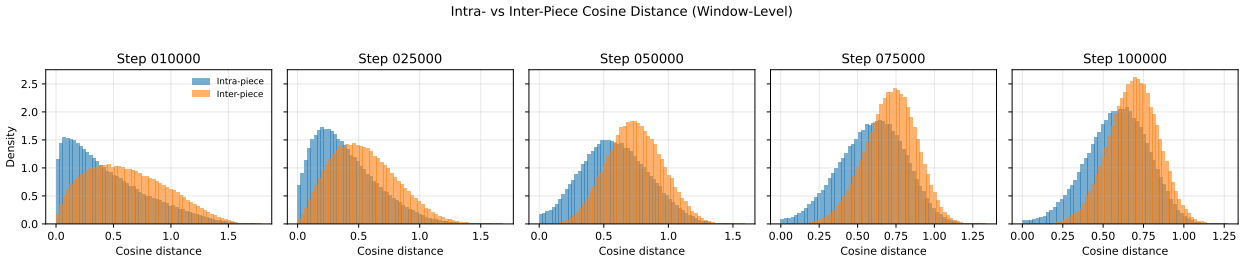


Figure 5: Window-level cosine distance distributions for intra-piece (same piece) and inter-piece (different pieces) pairs across training checkpoints. Distributions increasingly overlap with training, indicating reduced piece-level separation.

Our results show that JEPA-style training progressively sharpens temporal representations while promoting increasingly isotropic and high-dimensional embeddings. At the same time, improved uniformity is accompanied by reduced piece-level separability, and downstream classification performance peaks at intermediate stages rather than improving monotonically. These findings point to a tradeoff between predictability, global uniformity, and discriminative structure in predictive self-supervised learning.

This work complements prior JEPA studies by emphasizing representation diagnostics over downstream benchmarks and by making explicit the geometric consequences of SIGReg regularization in an audio setting. Future work may extend this analysis to larger-scale datasets, alternative predictive objectives, or token-level temporal modeling to further investigate how architectural and regularization choices influence representation structure.

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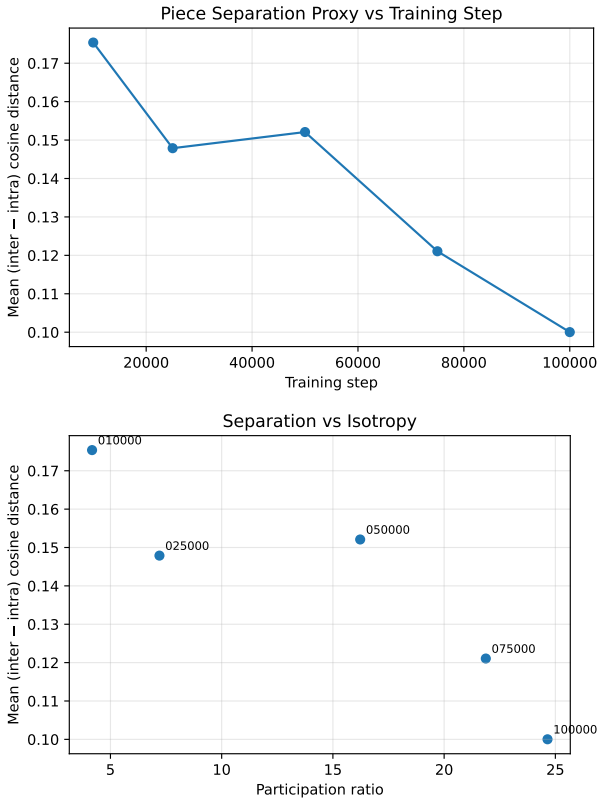


Figure 6: Relationship between piece-level separation and representation geometry. *Top*: mean inter–intra cosine distance gap versus training step. *Bottom*: separation versus isotropy, illustrating a tradeoff between uniformity and discriminability.